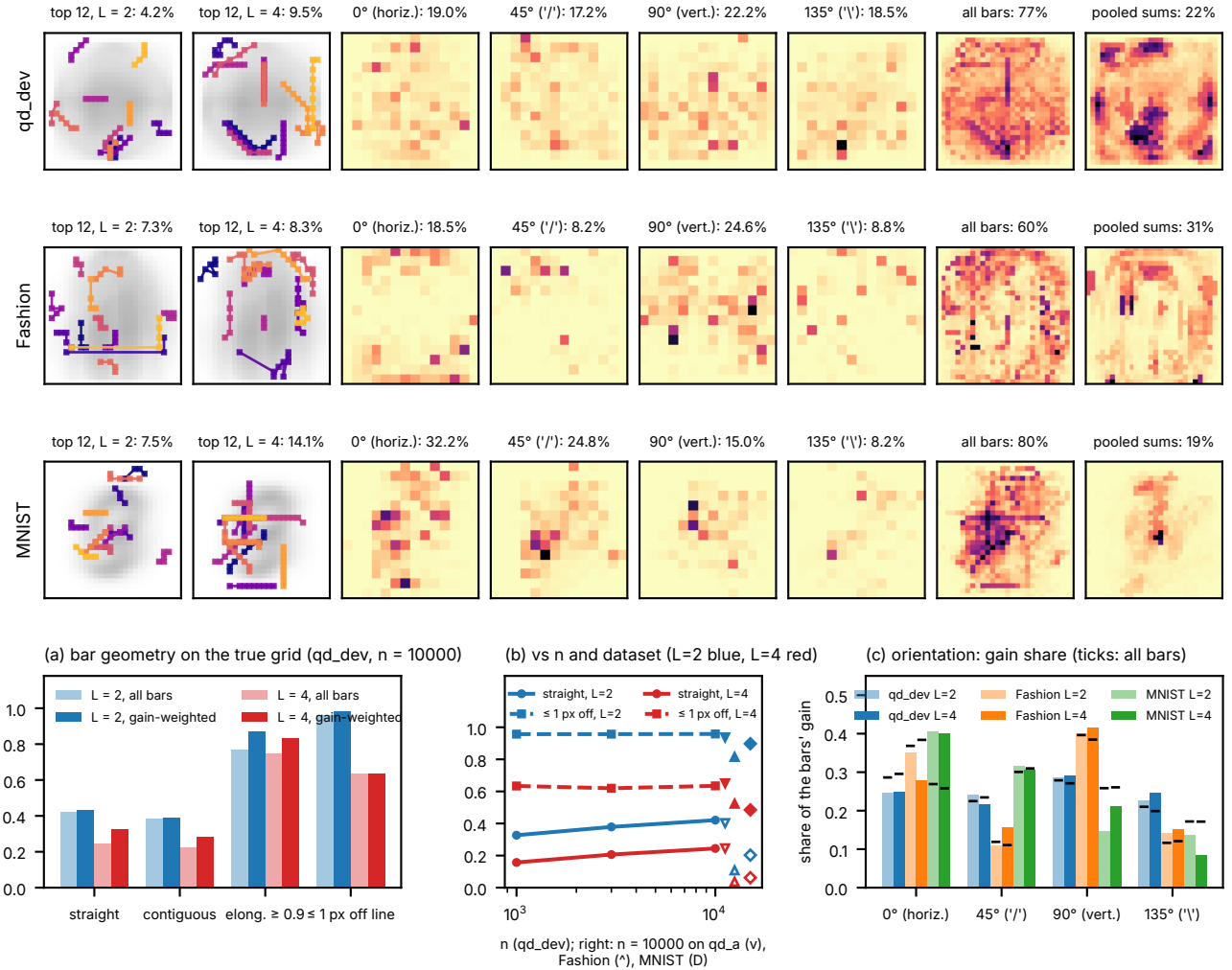

qd_dev
top 12, L = 2: 4.2%
top 12, L = 4: 9.5%
0° (horiz.): 19.0%
45° ('/'): 17.2%
90° (vert.): 22.2%
135° ('\'): 18.5%
all bars: 77%
pooled sums: 22%
Fashion
top 12, L = 2: 7.3%
top 12, L = 4: 8.3%
0° (horiz.): 18.5%
45° ('/'): 8.2%
90° (vert.): 24.6%
135° ('\'): 8.8%
all bars: 60%
pooled sums: 31%
MNIST
top 12, L = 2: 7.5%
top 12, L = 4: 14.1%
0° (horiz.): 32.2%
45° ('/'): 24.8%
90° (vert.): 15.0%
135° ('\'): 8.2%
all bars: 80%
pooled sums: 19%
(a) bar geometry on the true grid (qd_dev, n = 10000)
L = 2, all bars
L = 2, gain-weighted
L = 4, all bars
L = 4, gain-weighted
1.0
0.8
0.6
0.4
0.2
0.0
straight
contiguous
elong. ≥ 0.9
≤ 1 px off line
(b) vs n and dataset (L=2 blue, L=4 red)
straight, L=2
straight, L=4
≤ 1 px off, L=2
≤ 1 px off, L=4
1.0
0.8
0.6
0.4
0.2
0.0
10^3
10^4
n (qd_dev); right: n = 10000 on qd_a (v),
Fashion (^), MNIST (D)
(c) orientation: gain share (ticks: all bars)
share of the bars' gain
qd_dev L=2
qd_dev L=4
Fashion L=2
Fashion L=4
MNIST L=2
MNIST L=4
0.5
0.4
0.3
0.2
0.1
0.0
0° (horiz.)
45° ('/')
90° (vert.)
135° ('\')